Week 4 Tutorial Notes

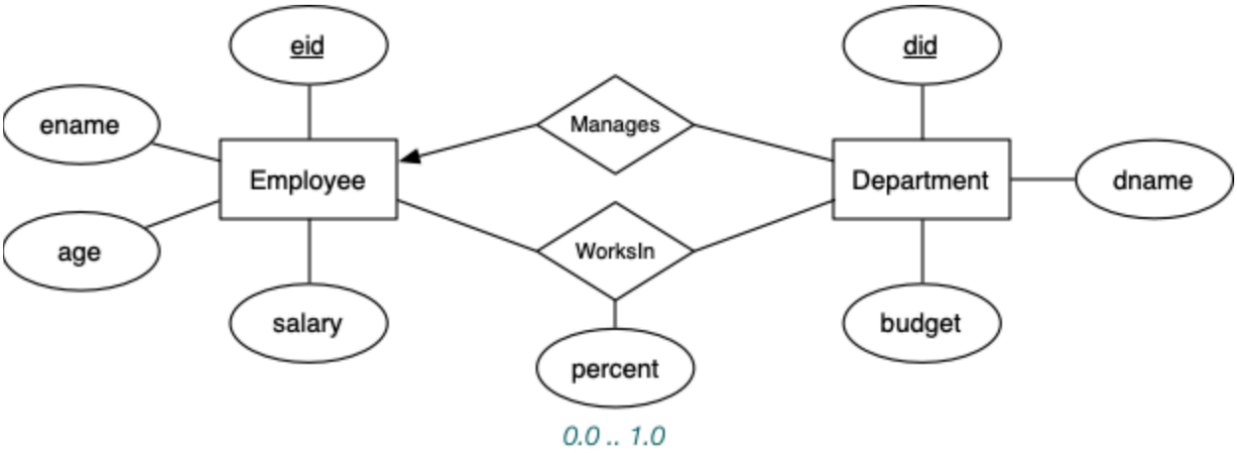
▼ Agenda

- Q1-10 → Constraints on create tables
- Q11 → SQL DELETE
- Q12-23 → SQL SELECT

Pre-Tute

- Quiz 3 due on Friday
- Assignment 1 due next Friday (Week 5)
- Help session... TBD

SQL Data Definition Language - a recap



▼ SQL CREATE TABLE

```
create table Employees (
      eid     integer primary key,
      ename   text,
      age     integer,
      salary  real,
      primary key (eid)
);
create table Departments (
      did     integer primary key,
      dname   text,
      budget  real,
      manager integer references Employees(eid)
);
create table WorksIn (
      eid     integer references Employees(eid),
      did     integer references Departments(did),
      percent real,
      primary key (eid,did)
);
```

SQL ALTER TABLE

- Changing column names

```
ALTER TABLE Departments
ALTER COLUMN dname department_name;
```

- Dropping columns

```
ALTER TABLE Departments
DROP COLUMN budget;
```

- Adding columns

```
ALTER TABLE Employees
ADD Email VARCHAR(255);
```

SQL Constraints

Describes the nature of an attribute/column's data:

- Data types

```
NUMERIC
INTEGER / INT
BOOLEAN

TEXT
CHAR(size)
VARCHAR(size)
ENUM(val1, val2, val3, ...)

DATE
TIME
DATETIME
```

- CHECK

```
CREATE TABLE Employees (
    age INTEGER CHECK (age >= 18 AND age <= 65),
    salary NUMERIC CHECK (salary > 0)
);
```

- NOT NULL

```
CREATE TABLE Students (
    student_id INTEGER NOT NULL,
```

```
    name VARCHAR(100) NOT NULL
);
```

▼ DEFAULT

```
CREATE TABLE Products (
    stock INT DEFAULT 0
);
```

▼ UNIQUE

```
CREATE TABLE Users (
    email VARCHAR(255) UNIQUE
);
```

▼ PRIMARY KEY

```
CREATE TABLE Books (
    bId CHAR(13) PRIMARY KEY = UNIQUE NOT NULL
);
```

▼ FOREIGN KEY

```
CREATE TABLE Orders (
    customer INT REFERENCES Customers(customer_id)
);
```

❓ Tutorial Q1-7

SQL DELETE

In a foreign key constraint, what happens to the child table when a referenced parent row is deleted?

- ON DELETE RESTRICT (or ON DELETE): Disallow the delete

```
CREATE TABLE Orders (
    customer_id INT,
    FOREIGN KEY (customer_id) REFERENCES Customers(id)
        ON DELETE RESTRICT
);
```

- ON DELETE CASCADE: Delete all child rows that refer to it

```
CREATE TABLE Orders (
    customer_id INT,
    FOREIGN KEY (customer_id) REFERENCES Customers(id)
        ON DELETE CASCADE
);
```

- ON DELETE SET DEFAULT: The foreign key in child rows are set to the default value (mostly commonly SET NULL)

```
CREATE TABLE Orders (
    customer_id INT DEFAULT 0,
    FOREIGN KEY (customer_id) REFERENCES Customers(id)
        ON DELETE SET DEFAULT
);
```

❓ Tutorial Q8-11

SQL SELECT

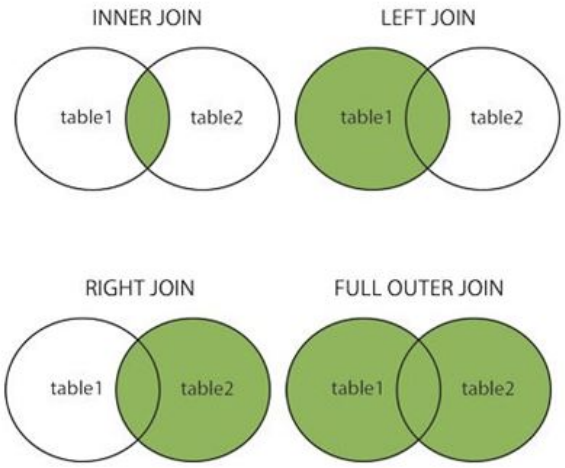
```
SELECT
    att1, att2, ...
FROM
    table_name
WHERE
    condition
<ORDER BY, GROUP BY>
```

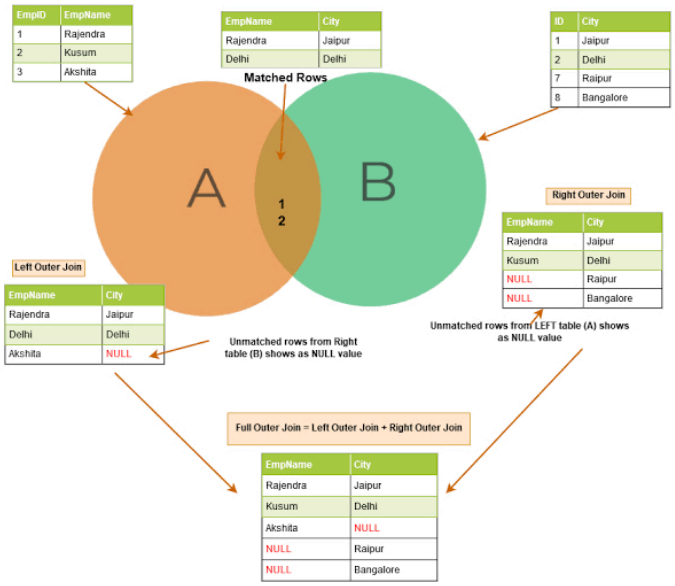
🛠️ To force unique output:

```
SELECT DISTINCT ...
```

SQL JOIN

```
SELECT
    att1, att2, ...
FROM
    tableA
JOIN
    tableB ON (tableA.id = tableB.Aid)
```





SQL Aggregates

A **SELECT** statement returns a table of tuples. We may want to aggregate them:

- count
- sum

- min
- max
- avg
- string_agg

▼ Example: Count of all employees with salary > 1000

```
SELECT
    count(eid)
FROM
    Employees
WHERE
    salary > 1000
;
```

▼ Example: Average salary

```
SELECT
    avg(salary)
FROM
    Employees
;
```

SQL GROUP BY

GROUP BY collapses all rows that share the same value for selected columns into a single row. Since multiple rows become one, any column you **SELECT** must either be:

- one of the grouped columns, or
- aggregated (eg. count, avg) to reduce the many values to one.

You may find the keyword **HAVING** used to filter groups in combination with **GROUP BY**. Please note that:

- **WHERE** filters rows before grouping, and
- **HAVING** filters groups (collapsed rows) after aggregation

▼ Example: **GROUP BY** used to group employees by salary, using **WHERE** and **HAVING**.

```
-- WHERE: filters individual rows before grouping
SELECT age, avg(salary)
FROM Employees
WHERE salary > 1000        -- removes rows first
GROUP BY age;

-- HAVING: filters the resulting groups
SELECT age, avg(salary)
FROM Employees
GROUP BY age
HAVING avg(salary) > 5000; -- removes groups after aggrega
ting
```

SQL CREATE VIEW

A view stores an SQL query. **NOTE:** views are not 'snapshots', they re-execute

```
CREATE VIEW view_name AS
SELECT
    *
FROM
    table_name
WHERE
```

```
    (conditions)
;
```